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Inventor(s)	Song; Il Jong

Relay device and electric vehicle charging controller comprising same

Abstract

A relay device according to an embodiment of the present invention comprises: a relay unit in which a switch operates according to a voltage applied to a coil; a first control unit that controls a voltage applied to the coil by turning on and off a first switching element according to a first control signal; a first discharge unit and a second discharge unit that discharge a back electromotive force generated in the coil when the relay unit is turned off; and a second control unit including a second switching element, and controlling the back electromotive force to be discharged through the first discharge unit or the second discharge unit by turning on and off the switching element according to a second control signal.

Inventors:	Song; Il Jong (Seoul, KR)
Applicant:	LG INNOTEK CO., LTD. (Seoul, KR)
Family ID:	77292504
Assignee:	LG INNOTEK CO., LTD. (Seoul, KR)
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Primary Examiner: Barnie; Rexford N

Assistant Examiner: Chowdhuri; Swarna N

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the U.S. national stage application of International Patent Application No. PCT/KR2021/001126, filed Jan. 28, 2021, which claims the benefit under 35 U.S.C. § 119 of Korean Application No. 10-2020-0015865, filed Feb. 10, 2020, the disclosures of each of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(2) Embodiments relate to a relay device and an electric vehicle charging controller including the same.

BACKGROUND ART

(3) Eco-friendly vehicles such as electric vehicles (EVs) or plug-in hybrid electric vehicles (PHEVs) use electric vehicle supply equipment (EVSE) installed in charging stations to charge batteries.

(4) To this end, an electric vehicle charging controller (EVCC) is equipped in the EV, communicates with the EV and the EVSE, and controls the charging of the electric vehicle.

(5) For example, when receiving a signal directing the start of charging from the electric vehicle, the EVCC may control the electric vehicle to start charging, and when receiving a signal directing the end of charging from the electric vehicle, the EVCC may control the electric vehicle to end charging.

(6) A charging method of the electric vehicle may be classified into fast charging and slow charging according to a charging time. In the fast charging, the battery is charged by a direct current (DC) supplied from a charger, and in the slow charging, the battery is charged by an alternating current (AC) supplied from the charger. Accordingly, the charger used for the fast charging is called a fast charger or a DC charger, and the charger used for the slow charging is called a slow charger or an AC charger.

(7) The electric vehicles are provided with a relay device to perform various operations such as charging of the battery or vehicle traveling. For example, the electric vehicles may also use the relay device to inhibit an inrush current from being generated in the battery or the like when a high voltage is applied to the electric vehicle supply equipment when charging the battery.

(8) The relay device controls the current by repeating an on-off state. However, there is a problem in that an accurate on-off control is difficult due to a back electromotive force generated in a relay coil upon turn-off. A relay discharge circuit is required to solve this problem.

Technical Problem

(9) An embodiment is directed to providing a relay device having high back electromotive force discharge performance and adaptability to a use environment.

(10) An embodiment is directed to providing an electric vehicle charging controller including a relay device having high back electromotive force discharge performance and adaptability to a use environment.

(11) An embodiment is directed to providing an electric vehicle having a relay device having high back electromotive force discharge performance and adaptability to a use environment.

(12) The objects of the embodiments are not limited to thereto and will also include objects or effects that may be identified from the configurations or embodiments to be described below.

Technical Solution

(13) A relay device according to an embodiment of the present invention includes: a relay unit in which a switch operates according to a magnitude of a voltage applied to a coil; a first control unit

configured to control the voltage applied to the coil by turning on and off a first switching element according to a first control signal; a first discharge unit and a second discharge unit configured to discharge a back electromotive force generated in the coil when the relay unit is turned off; and a second control unit including a second switching element, and configured to control the back electromotive force to be discharged through the first discharge unit or the second discharge unit by turning on and off the switching element according to a second control signal.

(14) The second control unit may control the back electromotive force generated in the coil when the first switching element is turned off to be discharged by the first discharge unit by turning off the second switching element.

(15) The second control unit may control the back electromotive force generated in the coil when the first switching element is turned off to be discharged by the second discharge unit by turning on the second switching element.

(16) The first control signal may be a pulse width modulation (PWM) signal.

(17) The first discharge unit may include: a first diode having an anode terminal connected to a first end of the coil; and a second diode having a cathode terminal connected to a cathode terminal of the first diode, and an anode terminal connected to a second end of the coil, a second end of the second switching element, and a first end of the first switching element.

(18) The second discharge unit may include a third diode having a cathode terminal connected to the first end of the coil, and an anode terminal connected to a first end of the second switching element.

(19) The first diode may be a Zener diode.

(20) A relay device according to another embodiment of the present invention includes: a switch; a coil disposed adjacent to the switch; a first diode having an anode terminal connected to a first end of the coil; a second diode having a cathode terminal connected to a cathode terminal of the first diode, and an anode terminal connected to a second end of the coil; a third diode having a cathode terminal connected to the first end of the coil; a first switching element having a first end connected to the anode terminal of the second diode, and a second end connected to a ground terminal; and a second switching element having a first end connected to an anode terminal of the third diode, and a second end connected to the anode terminal of the second diode.

(21) An electric vehicle charging controller according to still another embodiment of the present invention includes: a first signal detection unit connected to a first signal line and configured to detect a first charging sequence signal; a second signal detection unit connected to a second signal line and configured to detect a second charging sequence signal; a relay unit in which a switch operates according to a magnitude of a voltage applied to a coil; a first control unit configured to control the voltage applied to the coil by turning on and off a first switching element according to a first control signal; a first discharge unit and a second discharge unit configured to discharge a back electromotive force generated in the coil when the relay unit is turned off; and a second control unit including a second switching element, and configured to control the back electromotive force to be discharged through the first discharge unit or the second discharge unit by turning on and off the switching element according to a second control signal, wherein the coil is disposed between the first signal line and the second signal line, and has one end connected to a relay of a junction box.

(22) An electric vehicle according to yet another embodiment of the present invention includes: a relay unit in which a switch operates according to a magnitude of a voltage applied to a coil; a first control unit configured to control the voltage applied to the coil by turning on and off a first switching element according to a first control signal; a first discharge unit and a second discharge unit configured to discharge a back electromotive force generated in the coil when the relay unit is turned off; and a second control unit including a second switching element, and configured to control the back electromotive force to be discharged through the first discharge unit or the second discharge unit by turning on and off the switching element according to a second control signal, wherein the switch is disposed between a battery of the electric vehicle and an inverter configured

to drive a motor.

Advantageous Effects

(23) According to an embodiment, it is possible to provide a relay device adaptive to a driving situation.

(24) According to an embodiment, it is possible to provide a fast discharge of a back electromotive force generated in a relay coil.

(25) According to an embodiment, it is possible to greatly reduce an electromagnetic interference (EMI) noise when a back electromotive force generated in a relay coil is discharged.

(26) The diverse and beneficial advantages and effects of the present invention are not limited the above-described contents and will be understood more easily in a process of describing the specific embodiments of the present invention.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a view for describing an electric vehicle charging system according to an embodiment of the present invention.

(2) FIG. 2 is a view showing a configuration of the electric vehicle charging system according to the embodiment of the present invention.

(3) FIG. 3 is a view showing a circuit configuration of the electric vehicle charging system according to one embodiment of the present invention.

(4) FIG. 4 is a configuration diagram of a relay device according to the embodiment of the present invention.

(5) FIG. 5 is a circuit diagram of the relay device according to the embodiment of the present invention.

(6) FIG. 6 is a view showing a first driving example of the relay device according to the embodiment of the present invention.

(7) FIG. 7 is a view showing a second driving example of the relay device according to the embodiment of the present invention.

(8) FIG. 8 is a view showing a third driving example of the relay device according to the embodiment of the present invention.

(9) FIG. 9 is a view showing a fourth driving example of the relay device according to the embodiment of the present invention.

(10) FIGS. 10A to 10C show simulation results of the relay device according to the embodiment of the present invention.

(11) FIG. 11 is a view showing an electric vehicle charging controller according to the embodiment of the present invention.

(12) FIG. 12 is a view showing an electric vehicle according to the embodiment of the present invention.

MODES OF THE INVENTION

(13) Hereinafter, exemplary embodiments of the present invention will be described in detail with reference to the accompanying drawings.

(14) However, the technical spirit of the present invention is not limited to some embodiments described but may be implemented in various different forms, and one or more of the components may be used by being selectively coupled or substituted between the embodiments without departing from the technical spirit scope of the present invention.

(15) In addition, terms (including technical and scientific terms) used in the embodiments of the present invention may be construed as the meaning that may be generally understood by those skilled in the art to which the present invention pertains, unless specifically defined and described

explicitly, and the meaning of generally used terms such as terms defined in the dictionary may be construed in consideration of the contextual meaning of the related art.

(16) In addition, the terms used in the embodiments of the present invention are intended to describe the embodiments and are not intended to limit the present invention.

(17) In this specification, the singular form may also include the plural form unless otherwise specified in the phrase, and when it is described as “at least one (or one or more) of A and B, C”, it may include one or more of all possible combinations of A, B, and C.

(18) In addition, in describing the components of the embodiment of the present invention, terms such as first, second, A, B, (a), and (b) may be used.

(19) These terms are only intended to distinguish the component from other components, and the essence, sequence, or order of the corresponding components are not limited by the terms.

(20) In addition, when it is described that a component is ‘connected’, ‘coupled’, or ‘joined’ to another component, this may include a case in which the component is not only directly connected, coupled, or joined to another component, but also a case in which the component is ‘connected’, ‘coupled’, or joined to another component through other components interposed therebetween.

(21) In addition, when a component is described as being formed or disposed on “top (above) or bottom (below)” of each component, the top (above) or bottom (below) includes not only a case in which two components come into direct contact with each other but also a case in which one or more other components are formed or disposed between the two components. In addition, when expressed as “top (above) or bottom (below)”, this may also include the meaning of not only an upward direction but also a downward direction with respect to one component.

(22) FIG. 1 is a view for describing an electric vehicle charging system according to an embodiment of the present invention.

(23) An electric vehicle charging system according to an embodiment of the present invention may refer to a system for charging a battery of an electric vehicle operated by using electric energy as power.

(24) Referring to FIG. 1, the electric vehicle charging system according to the embodiment of the present invention may include electric vehicle supply equipment (EVSE) **10** and an electric vehicle (EV) **20**.

(25) The electric vehicle supply equipment **10** is equipment configured to supply alternating current (AC) or direct current (DC) power, may be disposed in a charging station or in a home, and may also be implemented to be portable. The electric vehicle supply equipment **10** may be used interchangeably with a supply, an AC supply, and a DC supply. The electric vehicle supply equipment **10** may receive the AC or DC power from a main power source. The main power source may include a power system or the like. The electric vehicle supply equipment **10** may transform or convert the AC or DC power supplied from the main power source to supply the transformed or converted power to the electric vehicle **20**.

(26) The electric vehicle **20** refers to a vehicle operated by receiving all or part of energy from an equipped battery. The electric vehicle **20** may include a plug-in hybrid electric vehicle (PHEV) that travels by using an engine using fossil fuel together as well as an electric vehicle that travels only with electric energy charged in the battery. The battery provided in the electric vehicle **20** may be charged by receiving power from the electric vehicle supply equipment **10**.

(27) FIG. 2 is a view showing a configuration of the electric vehicle charging system according to the embodiment of the present invention.

(28) The electric vehicle charging system according to the embodiment of the present invention may include the electric vehicle supply equipment (EVSE) **10**, a cable **50**, a connector **51**, an inlet **52**, a junction box **100**, an electric vehicle charging controller (EVCC) **200**, a battery **300**, a battery management system (BMS) **400**, and an electric power control unit (EPCU) **500**. A configuration included in the electric vehicle charging system may be classified into a configuration of the electric vehicle supply equipment **10** side (EVSE side) and a configuration of the electric vehicle

20 side (EV side). The configuration of the electric vehicle supply equipment **10** side may include the electric vehicle supply equipment **10**, the cable **50**, and the connector **51**. The configuration of the electric vehicle side may include the inlet **52**, the junction box **100**, the electric vehicle charging controller **200**, the battery **300**, the battery management system **400**, and the electric power control unit **500**. The classification is for convenience of description and is not limited thereto.

(29) First, the electric vehicle supply equipment **10** supplies power for charging the battery **300** of the electric vehicle. The electric vehicle supply equipment **10** may transmit power supplied from the main power source (e.g., the power system) to the electric vehicle **20**. At this time, the electric vehicle supply equipment **10** may reduce or convert the power supplied from the main power source to the electric vehicle **20**. In one embodiment, when the electric vehicle supply equipment **10** supplies AC power to the electric vehicle **20**, the electric vehicle supply equipment **10** may transform the AC power supplied from the main power source to supply the transformed AC power to the electric vehicle **20**. In another embodiment, when the electric vehicle supply equipment **10** supplies DC power to the electric vehicle **20**, the electric vehicle supply equipment **10** converts the AC power supplied from the main power source into DC power to supply the DC power to the electric vehicle **20**. To transform or convert power, the electric vehicle supply equipment **10** may be provided with a power conversion system. In the embodiment, the electric vehicle supply equipment **10** may include a rectifier, an isolation transformer, an inverter, a converter, and the like.

(30) The electric vehicle supply equipment **10** may include a charging control device configured to transmit and receive various control signals necessary for charging the battery **300** of the electric vehicle **20** and control a battery charging process. The charging control device may transmit and receive a control signal to and from the electric vehicle **20** and perform the battery charging process. The control signal may include information such as ready to charge, end of charge, and proximity detection. The charging control device may include a communication device configured to communicate with the electric vehicle **20**. The communication device may communicate with the electric vehicle **20** using power line communication (PLC), a controller area network (CAN), or the like. The communication device may also be included in the charging control device or may also be configured separately.

(31) Next, the cable **50**, the connector **51**, and the inlet **52** electrically connect the electric vehicle supply equipment **10** and the electric vehicle.

(32) The cable **50** transfers power and signals between the electric vehicle supply equipment **10** and the electric vehicle **20**. The cable **50** may include a power line transmitting power, a signal line transmitting a control signal related to charging, a ground line connecting a ground, and the like.

(33) The cable **50** is connected to the electric vehicle supply equipment **10**. In the embodiment, the electric vehicle supply equipment **10** and the cable **50** may be directly connected without a separate connection configuration. In another embodiment, the electric vehicle supply equipment **10** and the cable **50** may be connected through a combination of a socket-outlet provided in the electric vehicle supply equipment **10** and a plug provided in the cable **50**.

(34) The connector **51** may be connected to the cable **50**, and the inlet **52** may be provided in the electric vehicle **20**. The connector **51** and the inlet **52** may be bundled together to be referred to as a coupler. The connector **51** and the inlet **52** have a structure that may be coupled to each other, and the electric vehicle **20** and the electric vehicle supply equipment **10** may be electrically connected through the coupling between the connector **51** and the inlet **52**. The inlet **52** and the connector **51** may be connected not only directly, but also through an adapter.

(35) The connector **51** and the inlet **52** may be provided with a plurality of pins that may be coupled to each other. For example, one of the plurality of pins may be a pin for a CP port through which a control pilot (CP) signal is transmitted between the electric vehicle supply equipment **10** and the electric vehicle charging controller **200**, another one may be a pin for a proximity detection (PD) port that detects the proximity of the connector **51** and the inlet **52**, and still another one may be a pin for a protective earth (PE) port connected to a protective ground of the electric vehicle

supply equipment **10**. Yet another one of the plurality of pins may be a pin for driving a motor configured to open a fuel filler flap, yet another one may be a pin for sensing a motor, yet another one may be a pin for sensing a temperature, yet another one may be a pin for sensing a light emitting diode (LED), and yet another one may be a pin for CAN communication. One of the plurality of pins may be a pin for a voltage line applied from a collision detection sensor in the electric vehicle **20**, another one may be a battery pin for supplying charging power to the electric vehicle **20**, and still another one may be a pin for high voltage protection. However, the number and functions of pins are not limited thereto, and may be variously modified.

(36) The junction box **100** transmits the power supplied from the electric vehicle supply equipment **10** to the battery **300**. The power supplied from the electric vehicle supply equipment **10** is a high voltage, and when the high voltage is directly supplied to the battery **300**, the battery **300** may be damaged due to an inrush current. The junction box **100** may include at least one relay in order to inhibit damage to the battery due to the inrush current.

(37) The electric vehicle charging controller **200** may control a part or all of the process related to charging the battery of the electric vehicle **20**. The electric vehicle charging controller **200** may be referred to as an electric vehicle communication controller (EVCC).

(38) The electric vehicle charging controller **200** may communicate with the electric vehicle supply equipment **10**. The electric vehicle charging controller **200** may transmit and receive a control command related to the battery charging process to and from the electric vehicle supply equipment **10**. In one embodiment, the electric vehicle charging controller **200** may communicate with a charge control device provided in the electric vehicle supply equipment **10**, and may transmit and receive the control command related to the battery charging process to and from the charge control device.

(39) The electric vehicle charging controller **200** may communicate with the electric vehicle **20**. The electric vehicle charging controller **200** may receive the control command related to the battery charging process from the electric vehicle **20**. In one embodiment, the electric vehicle charging controller **200** may communicate with the battery management system **400** of the electric vehicle **20**, and also receive the control command related to the battery charging process from the battery management system **400**. In another embodiment, the electric vehicle charging controller **200** may communicate with the electric power control unit **500** of the electric vehicle **20**, and receive the control command related to the battery charging process from the electric power control unit **500**.

(40) The electric vehicle charging controller **200** may include a micro controller unit (MCU), a communication device, a relay device, and the like in order to perform the above function.

(41) The battery management system **400** manages an energy state of the battery **300** in the electric vehicle **20**. The battery management system **400** may monitor a usage status of the battery **300** and perform a control for efficient energy distribution. For example, the battery management system **400** may transmit an available power status of the electric vehicle **20** to a vehicle control unit and an inverter for efficient use of energy. As another example, the battery management system **400** may drive a cooling fan to correct a voltage deviation for each cell of the battery **300** or maintain the battery **300** at an appropriate temperature.

(42) The electric power control unit **500** is a unit configured to control the overall movement of the electric vehicle, including the control of the motor. The electric power control unit **500** may include a motor control unit (MCU), a low voltage DC-DC converter (LDC), and a vehicle control unit (VCU). The motor control unit may be referred to as an inverter. The motor control unit may receive DC power from the battery to convert the DC power into three-phase AC power, and control the motor according to a command of the vehicle control unit. The low voltage DC-DC converter may convert high voltage power into low voltage (e.g., 12 [V]) power to supply the low voltage power to each part of the electric vehicle **20**. The vehicle control unit functions to maintain the performance of the system with respect to the electric vehicle **20** as a whole. The vehicle control unit may perform various functions such as charging and vehicle traveling together with

various devices such as the motor control unit and the battery management system **400**.

(43) FIG. **3** is a view showing a circuit configuration of the electric vehicle charging system according to one embodiment of the present invention.

(44) Referring to FIG. **3**, the electric vehicle charging system according to the embodiment of the present invention includes the electric vehicle supply equipment **10**, the connector **51**, the inlet **52**, and the electric vehicle **20**.

(45) First, the electric vehicle supply equipment **10** may include overload breakers RCBO**1** and RCBO**2**, a power conversion system PCS, an insulation monitoring device CT, a communication device COM**1**, a plurality of power lines DC+ and DC−, a plurality of signal lines C**1** to C**6**, and a ground line FE. The plurality of power lines DC+ and DC−, the plurality of signal lines C**1** to C**6**, and the ground line FE may extend to the electric vehicle **20** through the coupling between the connector **51** and the inlet **52**.

(46) The electric vehicle supply equipment **10** may receive AC power from the power system. The received AC power may pass through the overload breakers RCBO**1** and RCBO**2**. The overload breakers RCBO**1** and RCBO**2** may function to block the reception of the AC power when the electric vehicle supply equipment **10** is overloaded.

(47) The AC power passing through the overload breaker RCBO**1** is input to the power conversion system PCS, and converted into DC power. The power conversion system PCS supplies the DC power to the electric vehicle **20** through two power lines DC+ and DC−. A diode a configured to block a reverse voltage from the electric vehicle **20** may be disposed on a first power line DC+ of the two power lines DC+ and DC−, and a fuse u configured to inhibit damage due to an overvoltage applied from the electric vehicle **20** may be disposed on a second power line DC−.

(48) The insulation monitoring device CT may be disposed between the two power lines DC+ and DC− and the ground. The insulation monitoring device CT may monitor an insulation status of the two power lines DC+ and DC−.

(49) A first signal line C**1** and a second signal line C**2** may mean signal lines representing a start/stop state of the electric vehicle supply equipment **10**. The first signal line C**1** and the second signal line C**2** may transmit charging sequence signals, such as ready to charge and end of charge, from the electric vehicle supply equipment **10** to the electric vehicle **20**. To this end, a power of 12 [V] may be connected to one end of the first signal line C**1**, and the ground may be connected to one end of the second signal line C**2**. In addition, two switch devices d**1** and d**2** may be disposed on the first signal line C**1** and the second signal line C**2**, respectively. In the electric vehicle supply equipment **10**, the two switch devices d**1** and d**2** may transmit the charging sequence signal to the electric vehicle through an on-off operation.

(50) A third signal line C**3** may mean a signal line representing a connection state between the connector **51** and the inlet **52**. The third signal line C**3** may transmit a proximity signal according to the connection state between the connector **51** and the inlet **52**. One end of the third signal line C**3** may be connected to the second signal line C**2**.

(51) A fourth signal line C**4** may mean a signal line for approving charging permission for the electric vehicle **20**. The fourth signal line C**4** may transmit a control signal such as charge start or charge stop from the electric vehicle **20** to the electric vehicle supply equipment **10**. The fourth signal line C**4** may be connected to a signal detection device j, and the signal detection device j may detect a control signal transmitted through the fourth signal line C**4**.

(52) A fifth signal line C**5** and a sixth signal line C**6** may mean signal lines for data communication. The fifth signal line C**5** and the sixth signal line C**6** may be connected to the communication device COM**1**.

(53) Next, the electric vehicle may include the junction box **100**, the electric vehicle charge controller **200**, and the battery **300**. The electric vehicle **20** may include the plurality of power lines DC+ and DC−, the plurality of signal lines C**1** to C**6**, and the ground line FE.

(54) The junction box **100** may be connected to the two power lines DC+ and DC−. The junction

box **100** may include two contactors **c** disposed on each of the two power lines DC+ and DC-. The two contactors may be turned on and off by the electric vehicle charging controller **200**. The junction box **100** may be connected to the battery **300** through the two power lines DC+ and DC-, and may transmit the DC power received from the electric vehicle supply equipment **10** to the battery **300** to perform charging.

(55) The electric vehicle charging controller **200** may include a relay device **e**, a plurality of signal detection devices **f**, **g**, and **h**, a switch **k**, and a communication device **COM2**. The electric vehicle charging controller **200** may be connected to the plurality of signal lines **C1** to **C6** and the ground line **FE**.

(56) The relay device **e** may be disposed between the first signal line **C1** and the second signal line **C2**. Specifically, one end of the relay device **e** may be connected to the second signal line **C2**, and the other end thereof may be connected to the first signal line **C1**. At this time, the two contactors **c** may be connected between the other end of the relay device **e** and the first signal line **C1**. The relay device **e** may control the opening and closing of the two contactors **c** through an opening/closing operation.

(57) A first signal detection device **f** and a second signal detection device **g** are connected to the first signal line **C1** and the second signal line **C2**, respectively. The two signal detection devices **f** and **g** may detect a signal generated when the two switch devices **d1** and **d2** provided in the electric vehicle supply equipment **10** are turned on. The two signal detection devices **f** and **g** may transmit the detected signal to the micro controller, the vehicle control unit, or the like included in the electric vehicle charging controller **200**.

(58) A third signal detection device **h** is connected to the third signal line **C3**. The third signal detection device **h** may detect a signal for detecting the connection state between the connector **51** and the inlet **52**.

(59) The switch **k** is connected to the fourth signal line **C4**. When the switch **k** is turned on, a signal indicating the charge start may be transmitted to the electric vehicle supply equipment **10**.

(60) The communication device **COM2** is connected to the fifth signal line **C5** and the sixth signal line **C6**. The communication device **COM2** may communicate with the communication device **COM1** through the fifth signal line **C5** and the sixth signal line **C6**.

(61) FIG. **4** is a configuration diagram of a relay device according to the embodiment of the present invention.

(62) The relay device according to the embodiment of the present invention may include a relay unit **610**, a first control unit **620**, a first discharge unit **630**, a second discharge unit **640**, and a second control unit **650**.

(63) The relay unit **610** includes a switch and a coil. The relay unit **610** operates the switch according to the magnitude of a voltage applied to the coil. When the voltage is cut off while being supplied, a back electromotive force may be generated in the coil.

(64) The first control unit **620** controls the voltage applied to the coil by turning on and off a first switching element according to a first control signal. The first control signal may be a pulse width modulation (PWM) signal. Accordingly, the first switching element periodically repeats turn-on and turn-off operations.

(65) The first discharge unit **630** and the second discharge unit **640** discharge the back electromotive force generated in the coil when the relay unit **610** is turned off. The first discharge unit **630** and the second discharge unit **640** do not simultaneously discharge the back electromotive force generated in the coil. In other words, when the first discharge unit **630** discharges the back electromotive force, the second discharge unit **640** does not discharge the back electromotive force, and when the second discharge unit **640** discharges the back electromotive force, the first discharge unit **630** does not discharge the back electromotive force.

(66) The second control unit **650** includes a second switching element and controls the back electromotive force to be discharged through the first discharge unit **630** or the second discharge

unit **640** by turning on and off the switching element according to a second control signal. In other words, the second control unit **650** may select a discharge unit to discharge the back electromotive force generated in the coil.

(67) Specifically, the second control unit **650** may control the back electromotive force generated in the coil when the first switching element is turned off to be discharged by the first discharge unit **630** by turning off the second switching element.

(68) In addition, the second control unit **650** may control the back electromotive force generated in the coil when the first switching element is turned off to be discharged by the second discharge unit **640** by turning on the second switching element.

(69) FIG. 5 is a circuit diagram of the relay device according to the embodiment of the present invention.

(70) The relay unit **610** includes a switch SW and a coil L. The coil L has a magnetic property when a current is applied, and may perform an opening/closing operation by pulling a switch SW made of iron. A first end of the coil L may be connected to a voltage source VCC, and a second end of the coil L may be connected to a first switching element Q1. The switch SW may be connected to one end of a circuit such as an electric vehicle or electric vehicle supply equipment.

(71) The first control unit **620** includes the first switching element Q1 and a first switching control circuit CON1.

(72) The first switching element Q1 may be a metal oxide semiconductor field effect transistor (MOSFET). The first switching element Q1 may include a drain terminal, a source terminal, and a gate terminal. The drain terminal of the first switching element Q1 may be connected to the first discharge unit **630** and the second control unit **650**. The source terminal of the first switching element Q1 may be connected to a ground terminal. The gate terminal of the first switching element Q1 may be connected to the first switching control circuit CON1.

(73) The first switching control circuit CON1 controls the opening and closing of the first switching element Q1 according to a received first control signal S1. The first switching control circuit CON1 may control the opening and closing of the first switching element Q1 by applying a voltage to the gate terminal of the first switching element Q1 according to the first control signal S1.

(74) The first discharge unit **630** includes a first diode D1 and a second diode D2.

(75) An anode terminal of the first diode D1 is connected to the first end of the coil. A cathode terminal of the first diode D1 is connected to a cathode terminal of the second diode D2. The first diode D1 may be a Zener diode.

(76) The cathode terminal of the second diode D2 is connected to the cathode terminal of the first diode D1. An anode terminal of the second diode D2 is connected to the second end of the coil L. The anode terminal of the second diode D2 is connected to the first end of the first switching element Q1. The anode terminal of the second diode D2 may be connected to the drain terminal of the first switching element Q1. The anode terminal of the second diode D2 is connected to a second end of a second switching element Q2. The anode terminal of the second diode D2 may be connected to an emitter terminal of the second switching element Q2.

(77) The second discharge unit **640** may include a third diode D3.

(78) A cathode terminal of the third diode D3 is connected to the first end of the coil L. The cathode terminal of the third diode D3 may be connected to the anode terminal of the first diode D1. An anode terminal of the third diode D3 is connected to a first end of the second switching element Q2. The anode terminal of the third diode D3 may be connected to a collector terminal of the second switching element Q2.

(79) The second control unit **650** includes the second switching element Q2 and a second switching control circuit CON2.

(80) The second switching element Q2 may be a bipolar junction transistor (BJT). The second switching element Q2 may include a collector terminal, an emitter terminal, and a base terminal.

The collector terminal of the second switching element Q2 may be connected to the anode terminal of the third diode D3. The emitter terminal of the second switching element Q2 may be connected to the second end of the coil L. The emitter terminal of the second switching element Q2 may be connected to the anode terminal of the second diode D2. The emitter terminal of the second switching element Q2 may be connected to the first control unit 620. The base terminal of the second switching element Q2 may be connected to the second switching control circuit CON2. (81) The second switching control circuit CON2 controls the opening and closing of the second switching element Q2 according to a received second control signal S2. The second switching control circuit CON2 may control the opening and closing of the second switching element Q2 by applying a voltage to the base terminal of the second switching element Q2 according to the second control signal S2.

(82) Next, a driving process of the relay device according to the embodiment of the present invention will be described in detail with reference to FIGS. 6 to 10C.

(83) FIG. 6 is a view showing a first driving example of the relay device according to the embodiment of the present invention.

(84) In the first driving example, the first switching control circuit CON1 of the relay device 600 turns on the first switching element Q1 according to the first control signal S1, and the second switching control circuit CON2 turns on the second switching element Q2 according to the second control signal S2. Then, a current I1 flows according to a potential difference between the applied voltage source VCC and the ground. The current I1 passes through the coil L, and the coil L has the magnetic property by the current I1. The coil L having the magnetic property may turn on the switch SW. At this time, no current may flow in the first discharge unit 630 and the second discharge unit 640.

(85) FIG. 7 is a view showing a second driving example of the relay device according to the embodiment of the present invention.

(86) In the second driving example, the first switching control circuit CON1 of the relay device 600 turns off the first switching element Q1 according to the first control signal S1, and the second switching control circuit CON2 turns on the second switching element Q2 according to the second control signal S2. Since the first switching element Q1 is turned off, an electrical connection between the applied voltage source VCC and the ground is cut off. Accordingly, a back electromotive force is generated in the coil L. At this time, since the second switching element Q2 is in a state of being turned on, a current I2 due to the back electromotive force flows through a closed circuit formed by the coil L of the relay unit 610, the second discharge unit 640, and the second switching element Q2. Then, the back electromotive force may be discharged through the third diode D3 of the second discharge unit 640 in the closed circuit. At this time, the second discharge unit 640 may not be involved in the discharge of the back electromotive force.

(87) When the back electromotive force is discharged by the first discharge unit 630, the first diode D1 and the second diode D2 are involved in the reflux of the back electromotive force. In this case, the voltage of the coil L may be increased by the magnitudes of the voltages applied to the first diode D1 and the second diode D2. Accordingly, it is possible to quickly discharge the back electromotive force. However, an electromagnetic interference (EMI) noise may increase due to a sudden increase in the magnitude of the voltage.

(88) FIG. 8 is a view showing a third driving example of the relay device according to the embodiment of the present invention.

(89) In the third driving example, the first switching control circuit CON1 of the relay device 600 turns on the first switching element Q1 according to the first control signal S1, and the second switching control circuit CON2 turns off the second switching element Q2 according to the second control signal S2. Then, a current I3 flows according to the potential difference between the applied voltage source VCC and the ground. The current I3 passes through the coil L, and the coil L has the magnetic property by the current I3. The coil L having the magnetic property may turn on the

switch SW. At this time, no current may flow in the first discharge unit **630** and the second discharge unit **640**.

(90) FIG. **9** is a view showing a fourth driving example of the relay device according to the embodiment of the present invention.

(91) In the fourth driving example, the relay device turns off the first switching element **Q1** and turns off the second switching element **Q2**. Since the first switching element **Q1** is turned off, the electrical connection between the applied voltage source and the ground is cut off. Accordingly, the back electromotive force is generated in the coil. At this time, since the second switching element **Q2** is also in a state of being turned off, the current due to the back electromotive force does not flow to the second discharge unit **640**. The current due to the back electromotive force flows through the closed circuit formed by the coil of the relay unit **610** and the first discharge unit **630**. Then, the back electromotive force may be discharged through the first diode and the second diode of the second discharge unit **640** in the closed circuit. At this time, the first discharge unit **630** may not be involved in the discharge of the back electromotive force.

(92) When the back electromotive force is discharged by the second discharge unit **640**, the third diode **D3** is involved in the reflux of the back electromotive force. In this case, the voltage of the coil **L** may be increased by the magnitude of the voltage applied to the third diode **D3**. Since there is no sudden increase in the magnitude of the voltage, the electromagnetic interference (EMI) noise may be low. However, since only the third diode **D3** is involved in the reflux of the back electromotive force, a discharge speed of the back electromotive force may be slow.

(93) According to the first to fourth driving examples described with reference to FIGS. **6** to **9**, the discharge of the back electromotive force by the first discharge unit **630** and the discharge of the back electromotive force by the second discharge unit **640** have advantages and disadvantages opposite to each other. In consideration of this point, the relay unit **610** according to the embodiment of the present invention proposes a method of selecting a discharge method according to items required for a disposed environment. The relay device according to the embodiment of the present invention may operate according to the first driving example and the second driving example in an environment in which circuit stability needs to be enhanced by minimizing the occurrence of the EMI noise. On the other hand, in an environment in which a fast discharge of the back electromotive force is required, the relay device may operate according to the third and fourth driving examples.

(94) FIGS. **10A** to **10C** show simulation results of the relay device according to the embodiment of the present invention.

(95) FIG. **10A** shows the simulation result for the relay device including only the first discharge unit, FIG. **10B** shows the simulation result for the relay device including only the second discharge unit, and FIG. **10C** shows the simulation result for the relay device according to the embodiment of the present invention.

(96) TABLE-US-00001

TABLE 1	Case 1	Case 2	Case 3	Peak voltage (V)
40	23.5	23.5	when first switching element is turned on	Discharge time of 1
20.3	2.5	back electromotive force when first switching element is turned off		

(97) As shown in Table 1, in the relay device including only the first discharge unit, the peak voltage when the first switching element is turned on is 40 [V], and the discharge time of the back electromotive force when the first switching element is turned off is 1 [ms], so that it can be seen that the discharge time is short but the EMI noise is high due to the high peak voltage. In the relay device including only the second discharge unit, the peak voltage when the first switching element is turned on is 23.5 [V], and the discharge time of the back electromotive force when the first switching element is turned off is 20.3 [ms], so that it can be seen that the EMI noise is low due to the low peak voltage but the discharge time is slow. However, in the relay device according to the embodiment of the present invention, the peak voltage when the first switching element is turned on is 23.5 [V], and the discharge time of the back electromotive force when the first switching

element is turned off is 2.5 [ms], so that it can be seen that the EMI noise is low due to the low peak voltage and at the same time, fast discharge is possible. FIG. 11 is a view showing an electric vehicle charging controller according to the embodiment of the present invention. The electric vehicle charging controller **200** according to the embodiment of the present invention may transmit and receive signals to and from the electric vehicle supply equipment **10** through a plurality of signal lines and a ground line, and control the battery charging sequence using the transmitted and received signals.

(98) The electric vehicle charging controller **200** according to the embodiment of the present invention may include a first signal detection unit **f** and a second signal detection unit **g**. The first signal detection unit **f** is connected to a first signal line **C1** and detects a first charging sequence signal. The second signal detection unit **g** is connected to a second signal line **C2** and detects a second charging sequence signal. The first signal detection unit **f** detects the first charging sequence signal generated according to an on-off state of a first switch **d1** of the electric vehicle supply equipment **10**. The second signal detection unit **g** detects the second charging sequence signal generated according to an on-off state of a second switch **d2** of the electric vehicle supply equipment **10**.

(99) The electric vehicle charging controller **200** may include the relay device **600** according to the embodiment of the present invention. The relay device **600** may be disposed between the first signal detection unit **f** and the second signal detection unit **g**. The relay device **600** may be disposed between the first signal line **C1** and the second signal line **C2**. One end of the switch **SW** of the relay unit **610** included in the relay device **600** may be connected to the second signal line **C2**. The other end of the switch of the relay unit **610** included in the relay device **600** may be connected to the first signal line **C1**. At this time, the junction box **100** may be connected between the other end of the switch **SW** and the first signal line **C1**.

(100) The on and off operation of the switch **SW** is controlled by driving the first control unit **620**, the first discharge unit **630**, the second discharge unit **640**, and the second control unit **650** of the relay device **600**, and in the off state, the back electromotive force generated in the coil **L** is discharged. A relay of the junction box **100** may be repeatedly turned on and off according to the on-off operation of the relay device **600**. The relay device **600** according to the embodiment of the present invention functions to protect the battery from the high voltage power source of the electric vehicle supply equipment by turning on and off the relay of the junction box **100** by repeatedly turning on and off the relay unit **610**. At this time, the relay device **600** may stably control the operation of the junction box **100** by rapidly discharging the back electromotive force generated in the repeated on and off state of the relay unit **610** or by operating to minimize the occurrence of the EMI noise.

(101) FIG. 12 is a view showing an electric vehicle according to the embodiment of the present invention.

(102) The electric vehicle **20** according to the embodiment of the present invention may be driven by converting the power of the battery **300** to transmit the converted power to the motor by the electric power control unit **500**. More specifically, the electric vehicle **20** may be driven by converting the power of the battery **300** to transmit the converted power to the motor by the motor control unit (inverter) of the electric power control unit **500**.

(103) The electric vehicle **20** may include the relay device **600**. The relay device **600** may be disposed between the battery **300** and the electric power control unit **500**. One end of the switch **SW** of the relay unit **610** included in the relay device **600** is connected to the battery **300**, and the other end thereof is connected to the electric power control unit **500**. The on and off operation of the switch **SW** is controlled by driving the first control unit **620**, the first discharge unit **630**, the second discharge unit **640**, and the second control unit **650** of the relay device **600**, and in the off state, the back electromotive force generated in the coil **L** is discharged.

(104) The relay device **600** according to the embodiment of the present invention can inhibit

damage to a capacitor that is generated as the high voltage power of the battery is supplied to the motor control unit of the electric power control unit **500** by repeatedly turning on and off the relay unit **610**. The relay device **600** can improve the stability of the electric power control unit **500** by rapidly discharging the back electromotive force generated in the repeated on and off state of the relay unit or by operating to minimize the occurrence of the EMI noise.

(105) While the embodiments have been mainly described above, this is merely illustrative and does not limit the present invention, and those skilled in the art to which the present invention pertains will be able to understand that various modifications and applications not exemplified above are possible without departing from the essential characteristics of these embodiments. For example, each component specifically described in the embodiments may be modified and embodied. In addition, differences related to the modifications and applications should be construed as being included in the scope of the present invention defined in the appended claims.

Claims

1. A relay device comprising: a relay unit comprising a switch operating according to a magnitude of a voltage applied to a coil; a first control unit configured to control the voltage applied to the coil by turning on and off a first switching element according to a first control signal; a first discharge unit and a second discharge unit configured to discharge a back electromotive force generated in the coil when the relay unit is turned off; and a second control unit including a second switching element, wherein the second switching element is configured to be turned on or off based on a second control signal to select whether the back electromotive force generated in the coil is discharged through the first discharge unit or the second discharge unit.
2. The relay device of claim 1, wherein the second control unit is configured to control the back electromotive force generated in the coil when the first switching element is turned off to be discharged by the first discharge unit by turning off the second switching element.
3. The relay device of claim 1, wherein the second control unit is configured to control the back electromotive force generated in the coil when the first switching element is turned off to be discharged by the second discharge unit by turning on the second switching element.
4. The relay device of claim 1, wherein the first control signal is a pulse width modulation (PWM) signal.
5. The relay device of claim 1, wherein the first discharge unit includes: a first diode having an anode terminal connected to a first end of the coil; and a second diode having a cathode terminal connected to a cathode terminal of the first diode, and an anode terminal connected to a second end of the coil, a second end of the second switching element, and a first end of the first switching element.
6. The relay device of claim 5, wherein the second discharge unit includes a third diode having a cathode terminal connected to the first end of the coil, and an anode terminal connected to a first end of the second switching element.
7. The relay device of claim 5, wherein the first diode is a Zener diode.
8. A relay device comprising: a switch; a coil disposed adjacent to the switch; a first diode having an anode terminal connected to a first end of the coil; a second diode having a cathode terminal connected to a cathode terminal of the first diode, and an anode terminal connected to a second end of the coil; a third diode having a cathode terminal connected to the first end of the coil; a first switching element having a first end connected to the anode terminal of the second diode, and a second end connected to a ground terminal; and a second switching element having a first end connected to an anode terminal of the third diode, and a second end connected to the anode terminal of the second diode, wherein the second switching element is configured to be turned on

or off based on a control signal to select whether back electromotive force generated in the coil is discharged through the first diode and the second diode or the third diode.
